

Surface-Only Multiples: Mathematical Explanation

Converted from Lab1_calculations_surfaceMultiples.m

Overview

This document explains the simplified multiples algorithm where the downgoing reflection is restricted to the free surface (index $m = 0$). Indices n, p index buried reflecting interfaces; m is fixed to 0.

Reflection Signs

For a chosen pair (n, p) :

$$R_1 = R(n), \quad R_2 = -1 \quad (\text{surface}), \quad R_3 = R(p).$$

Layer and Segment Definitions

Define

$$\text{high} = \min(n, p), \quad \text{low} = \max(n, p).$$

Layer two-way time for layer i :

$$\Delta t_i = \frac{2 \Delta z_i}{v_p(i)}.$$

Three contiguous segments (indexed by sets of layer indices):

$$\mathcal{A} = \{1, \dots, \text{high}\}, \quad \mathcal{B} = \{\text{high} + 1, \dots, \text{low}\}, \quad \mathcal{C} = \{1, \dots, \text{high}\} \quad (\text{since } m = 0).$$

Segment two-way times:

$$\text{tw}_{\mathcal{A}} = \sum_{i \in \mathcal{A}} \Delta t_i, \quad \text{tw}_{\mathcal{B}} = \sum_{i \in \mathcal{B}} \Delta t_i, \quad \text{tw}_{\mathcal{C}} = \sum_{i \in \mathcal{C}} \Delta t_i.$$

Transmission and Attenuation

Transmission product for an index set $\mathcal{I}$ (excluding the reflecting interface where appropriate):

$$T(\mathcal{I}) = \prod_{j \in \mathcal{I}_T} (1 - R(j)^2).$$

Per-layer attenuation factor:

$$A_i = \exp\left(-\frac{32 \Delta t_i}{2 Q(i)}\right).$$

Segment attenuations:

$$Q_{\mathcal{A}} = \prod_{i \in \mathcal{A}} A_i, \quad Q_{\mathcal{B}} = \prod_{i \in \mathcal{B}} A_i, \quad Q_{\mathcal{C}} = \prod_{i \in \mathcal{C}} A_i.$$

Arrival Time and Amplitude

Total arrival two-way time for the surface multiple formed by (n, p) :

$$t_{\text{mult}} = \text{tw}_{\mathcal{A}} + \text{tw}_{\mathcal{B}} + \text{tw}_{\mathcal{C}}.$$

Discrete-time index chosen as the nearest sample to t_{mult} .

Amplitude without attenuation:

$$A_{\text{no-atten}} = R_1 R_2 R_3 T_{\mathcal{A}} T_{\mathcal{B}} T_{\mathcal{C}}.$$

With attenuation:

$$A_{\text{atten}} = A_{\text{no-atten}} Q_{\mathcal{A}} Q_{\mathcal{B}} Q_{\mathcal{C}}.$$

These amplitudes are accumulated into the multiple series at the computed time index.

Algorithmic Summary

Loop over $n = 1 \dots N$ and $p = 1 \dots N$ with $m = 0$:

1. Set $R_1, R_2 = -1, R_3$.
2. Form index sets $\mathcal{A}, \mathcal{B}, \mathcal{C}$ and compute T_*, tw_*, Q_* .
3. Compute t_{mult} and nearest time index; add $A_{\text{no-atten}}$ to M_1 and A_{atten} to M_2 .